medium. The performance of this capacitor may not be affected even it get heated.

ii) Variable Capacitor

These capacitors are used in radio receivers for tuning the receiver to a particular transmitting station. In this type, air is used as dielectric medium between Aluminium plates.

iii) Electrolytic Capacitor

Another type of capacitors, which can have a very large capacity of 10 to 100 μF in the electrolytic capacitor, is used in Radio circuit and Electric circuit. This is made by very thin Aluminium sheets which are separated by a thin layer of Aluminium Borate as Dielectric medium.

2.7.4 Uses of Capacitors

1. Capacitors in Fluorescent Tube

(i) When connected parallel with the supply, it improves power factor.

(ii) When connected in series with one of the two tube connected in parallel, it minimise the stroboscopic effect.

SRTY2H

(iii) It helps to avoid radio interference when connected parallel with the two contacts of tube starter. Here, it eliminates the sparkling due to the opening of two contacts.

2. In motors

When connected in series with starting winding of the single phase motor, it splits one phase into two phases and gives starting torque to the motor.

Example: Table fan, ceiling fan, and small one phase motors.

3. In parallel with the supply lines

To improve the power factor because it takes leading current.

4. In eliminators

(An appliance which reduces 230 v Ac to 3, 6, 9v Dc)

Capacitor is connected in parallel with Dc side to filter remaining A.C.

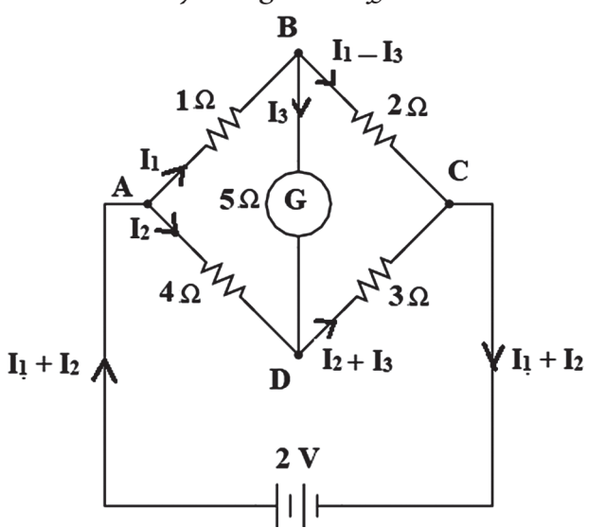
5. In petrol cars

It is connected in parallel with the two contacts of a distributor to avoid sparking due to opening of these contacts.

These are the uses of capacitors in various places.

Worked example:

Example:1 In an unbalanced bridge circuit, calculate the current flowing through all resistors by using Kirchoff's Law.



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